

Symbols, Names, and References

1 Computing with Symbols

- an example of a symbolic computation
- the symbolic ring

2 Names and References

- the assignment of names and values
- strings to prevent evaluation of symbols

MCS 320 Lecture 6
Introduction to Symbolic Computation
Jan Verschelde, 4 September 2026

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an example of a symbolic computation

$$\begin{aligned}(a + b\sqrt{2})(c + d\sqrt{2}) &= a(c + d\sqrt{2}) + b\sqrt{2}(c + d\sqrt{2}) \\&= ac + ad\sqrt{2} + b\sqrt{2}c + b\sqrt{2}d\sqrt{2} \\&= ac + ad\sqrt{2} + bc\sqrt{2} + bd2 \\&= (ac + 2bd) + (ad + bc)\sqrt{2}\end{aligned}$$

- The letters a , b , c , and d are symbols.
- As symbols, we compute with a , b , c , and d as with numbers.
- $a + b\sqrt{2}$ and $c + d\sqrt{2}$ are symbolic expressions, they are not algebraic numbers in $\mathbb{Q}(\sqrt{2})$.
- The last step in the calculation is a *normalization*.

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the Symbolic Ring

Why does $\sqrt{x^2}$ not automatically simplify to x ?

There are automatic simplifications, for example:

- $\sqrt{4}$ simplifies to 2
- $\sin(\pi)$ simplifies to 0

To prevent simplification in the examples above,

- 1 numbers must be treated as symbols, or be cast into the symbolic ring;
- 2 the function must be told to hold off the evaluation.

Holding off an evaluation is delaying the evaluation, till some point in the future when the hold is removed.

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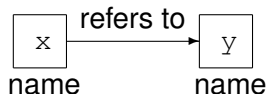
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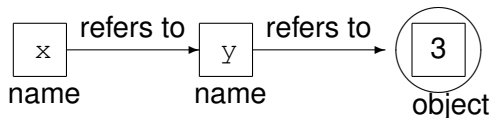
the Assignment of Names and Values

Let x and y both be the names of variables.

- ❶ After assigning y to x , we have the following picture:



- ❷ After assigning 3 to y , the situation is as follows:



The consequences of this arrangement:

- Evaluating y gives 3. Evaluating x to the fullest also gives 3.
- Assignments to y also change the value of x .

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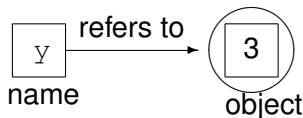
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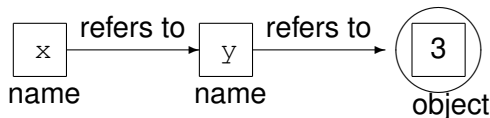
preventing the evaluation of symbols

Consider the following scenario.

- 1 We assigned 3 to the variable y , so we have:



- 2 Now, assign y to the variable x , so that this situation holds:



How do we prevent the evaluation of y to 3 , when we assign y to x ?